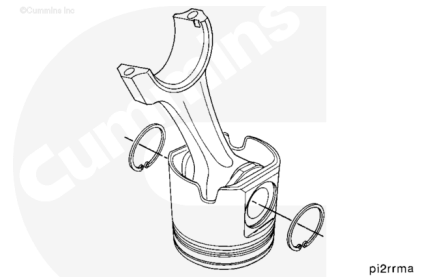


Trainings ▾

Remove

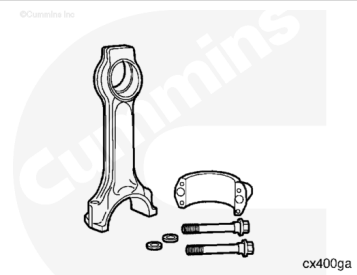
Remove the piston from the connecting rod. Refer to Procedure 001-054 in Section 1 (</qs3/pubsys2/xml/en/procedures/89/89-001-054.html>).



Clean

⚠ CAUTION ⚠

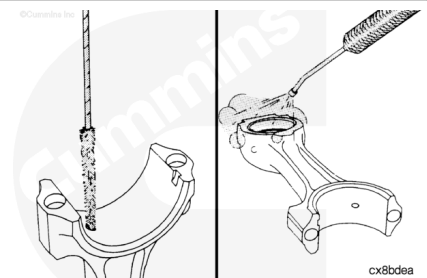
The number on the connecting rod must be the same as the number on the cap. Never assemble a new cap to an old rod or an old cap to a new rod.



Remove the capscrews and the cap from the connecting rod.

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use solvent or steam to clean the parts.

Dry with compressed air.

Use a soft bristle brush to clean the oil drilling.

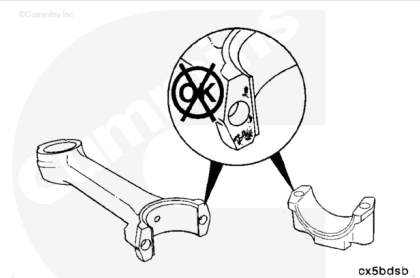
Dry the rod with compressed air.

Inspect for Reuse

Visually check the rod and the cap for fretting damage.

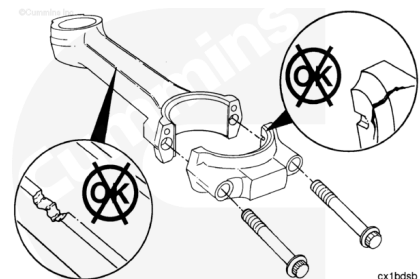
The rod and cap **must** be replaced as an assembly if fretting damage is visible on either piece.

Inspect for cracks and damage around the capscrew holes.



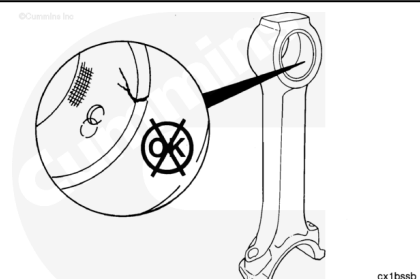
Inspect the rods and caps for damage.

Replace the rod and cap if either the I beam or cap is nicked or damaged.

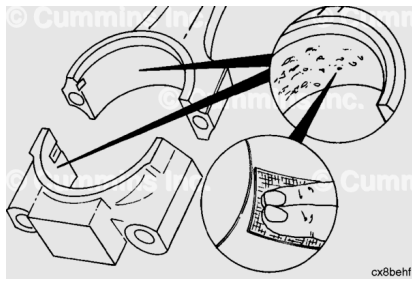


Visually inspect the rod pin bore bushing for damage or misalignment of the oil hole passage and bushing.

If the bushing is damaged the rod **must** be replaced.

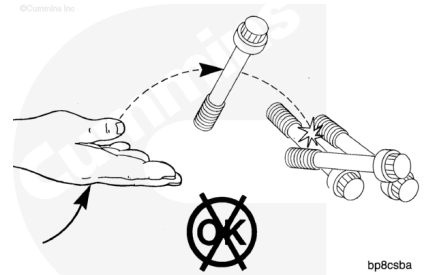


Visually inspect the bearing surface for nicks or burrs. If it is **not** possible to remove any nicks or burrs with a fine crocus cloth, the rod **must** be replaced.



⚠ CAUTION ⚠

Prevent damage to the capscrews. Nicks in the body of the capscrew can cause an area of stress that can fail during engine operation. Damage to the threads will cause torque values to be incorrect and will damage the mating parts.



Visually inspect the capscrew.

The capscrew **must** be replaced if:

- The threads are damaged
- Rust or corrosion has caused pitting in the body
- The body is nicked, galled, bent or stretched.

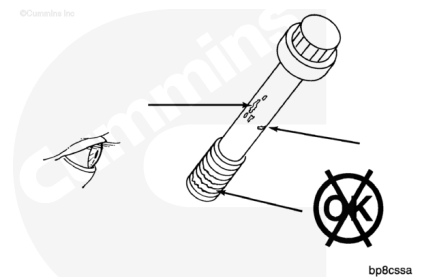
Note : Repair of rolled threads by the use of a thread die is **not recommended**. The thread die can create a sharp corner on the minor diameter (root) of the threads. This sharp corner can cause an area of increased stress.

Magnetic crack inspect the connecting rod capscrews.

Connecting rod capscrews **must** be magnetic crack inspected.

Refer to Procedure 017-011 in Section 17

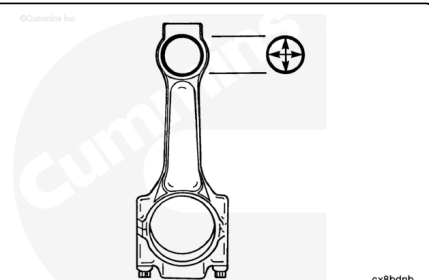
(/qs3/pubsys2/xml/en/procedures/89/89-017-011.html).



Measure

Measure the connecting rod bushing inside diameter.

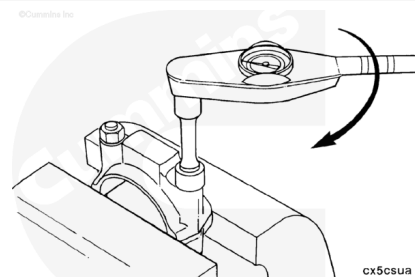
Connecting Rod Bushing I.D.		
mm		in
68.030	MIN	2.678
68.049	MAX	2.679



Note : The bushing is not replaceable without remanufacturing. If the bushing is not within specification the connecting rod must be replaced.

⚠ CAUTION ⚠

Use a vise with brass jaws to hold the connecting rod. Notches, scratches, or dents in the I-beam area could cause engine failure.



⚠ CAUTION ⚠

The number on the connecting rod must be the same number as the number on the rod cap. Never assemble a new cap to an old rod or an old cap to a new rod.

Note : Check the number of punch marks on the head of the mounting capscrews. If there already 4 punch marks, do not reuse the capscrew, replace it with a new part.

Use clean engine oil. Lubricate the connecting rod capscrews.

Assemble the connecting rod, the cap and the capscrews.

Note : Always use the torque-turn method to tighten the connecting rod capscrews when access permits.

Tighten the capscrews alternately and evenly to pull the cap over the dowel pins. Use the following steps and sequence to tighten the capscrews:

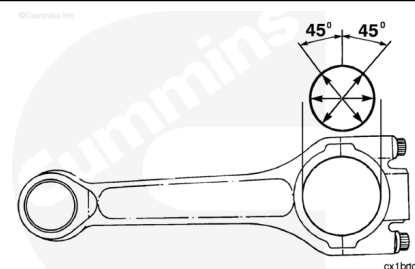
Torque Value:

1. 196 n•m [145 ft-lb]
2. Turn 90 degrees

Note : After tightening the capscrews make 1 punch mark on the head of each capscrew. Do not make a punch mark when using a new capscrew.

Use a dial bore indicator. Measure the I.D. using the measuring positions shown.

Connecting Rod Bearing Bore I.D.		
mm		in
115.000	MIN	4.5276
115.026	MAX	4.5286



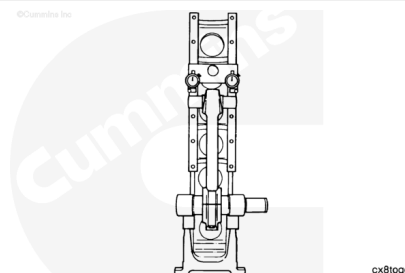
Note : If the measurements are **not** within specification the rod **must** be replaced or repaired.

Bend and Twist Inspect

Required tools:

- Part No. ST- 561 Connecting Rod Checking Fixture
- Part No. ST-1331 Connecting Rod Location Mandrel (for crank pin end of rod)
- Part No. 3165106 Connecting Rod locating mandrel (for piston rod end with bushing **installed**)

Bulletin No. 3377541, Service Tool Instructions - Connecting Rod Checking, is also available.



Calibration Procedure

⚠ CAUTION ⚠

The rod must be installed and torqued to specification or the measurement will be incorrect.

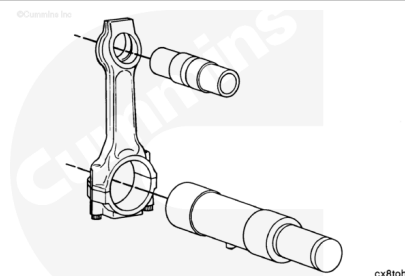
A connecting rod of known length, bend and twist, called a **reference** rod, is required to calibrate the fixture. A **new** connecting rod with a known distance from center of the crank pin to the centre of the piston pin end (rod length), can also be used.

Install the appropriate connecting rod mandrel into the reference connecting rod piston pin bore and center the mandrel.

Install the appropriate connecting rod mandrel into the reference connecting rod crankshaft bore.

Note : Make sure that the locating pin on the mandrel is down and centered in the connecting rod.

Lock the mandrel in position by rotating the end of the mandrel.



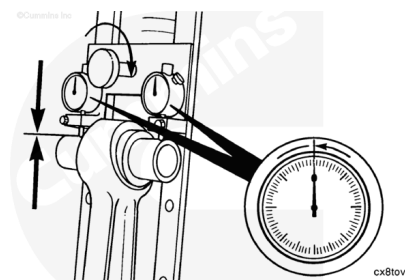
Install the reference connecting rod in the cranking fixture.

Loosen the knob and move the indicator bracket until both indicators touch the top of the piston pin end mandrel.

Move the bracket toward the mandrel until the indicator needles have moved approximately 0.25 mm (0.010 in).

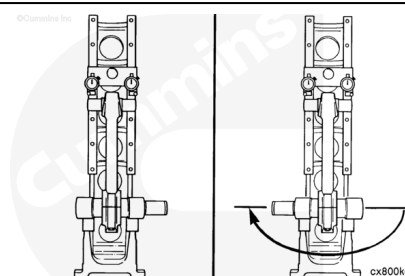
Turn the knob to tighten the bracket.

Adjust the indicator needles to "0". Move the connecting rod **in** and **out** to confirm the "0" setting.



Remove the reference connecting rod from the fixture.

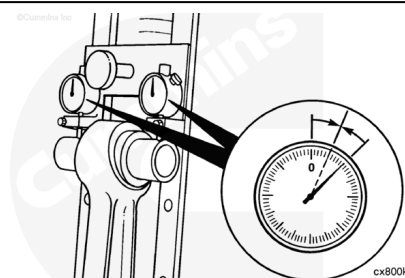
Rotate the connecting rod 180 degrees **horizontally** and install it in the checking fixture.



Be sure the indicators are "0" setting.

If the indicator needles do **not** return to the "0", adjust the indicator so that the "0" moves $\frac{1}{2}$ of the indicated difference from the needle to the "0" established during the previous step.

Note : If the needle is more than 0.10 mm [0.004 in] from the "0" established during the previous step, check for dirt or burrs on the mandrels and fixture. If the "0" is still **not** within specification, check to be sure that the **reference** connecting rod is **not** damaged.



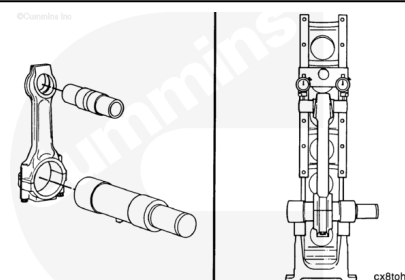
⚠ CAUTION ⚠

Do not adjust the indicators during this step of the procedure or the measurement will be incorrect.

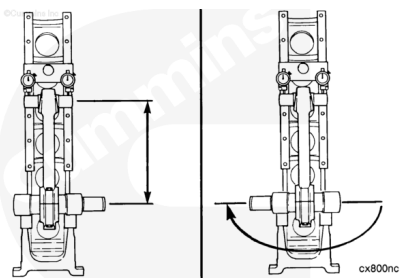
After completing the calibration of the fixture, remove the reference connecting rod.

Repeat the same steps to install the mandrels in the connecting rod to be measured, as you did for the reference rod.

Install the rod to be measured into the fixture.



Record the indicator reading. The difference of the calibrated "0" of the indicator **must** be added or subtracted from the known length of the reference rod to determine the length of the connecting rod being measured.



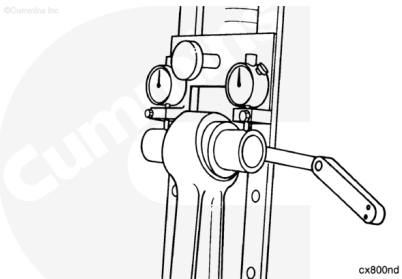
Connecting Rod Length		
mm		in
304.950	MIN	12.006
305.000	MAX	12.008

If the connecting rod length is **not** within specification, the connecting rod **must** be replaced.

Make a note of the indicator readings.

Remove the rod from the fixture. Turn the rod 180 degrees horizontally.

Compare the indicator readings with those noted in the previous step. The difference between the indicator readings is the amount of bend in the connecting rod.



Maximum Connecting Rod Bend (Bushing Installed)		
mm		in
0.10	MAX	0.004

If the connecting rod bend is **not** within specification, the connecting rod **must** be replaced.

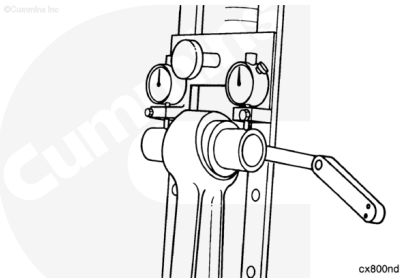
⚠ CAUTION ⚠

Never attempt to straighten a connecting rod with the use of heat or force. The connecting rod could eventually break and cause extensive engine damage.

Check the fixture and mandrel in the piston pin end for a gap between the two. If there is any twist in the connecting rod, the mandrel will **only** touch one side of the fixture.

Hold the end of the mandrel that is touching the fixture **firmly** against the fixture.

Use a feeler gauge to measure the gap between the mandrel and the fixture. the amount of gap between the mandrel and the fixture is the amount of connection rod twist.



Maximum Connecting Rod Twist (Bushing Installed)

mm		in
0.25	MAX	0.010

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If the connecting rod twist is **not** within specification, the connecting rod **must** be replaced.

⚠ CAUTION ⚠

Never attempt to straighten a connecting rod with the use of heat or force. The connecting rod could eventually break and cause extensive engine damage.

Magnetic Crack Inspect

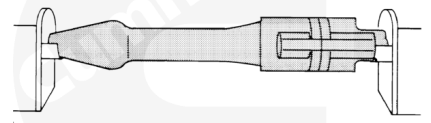
Use a magnetic particle testing machine.

The connecting rod and cap **must** be assembled during this check.

Use the residual method. Apply head shot amperage. Adjust the amperage to 1500 ampere D.C. or rectified A.C.

Check for cracks.

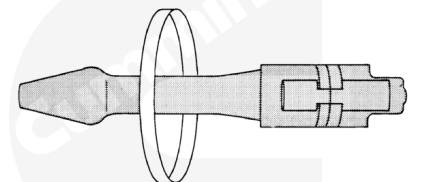
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cx8bdsc

Use the residual method. Apply coil shot amperage. Check for cracks.

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cx8bdsc

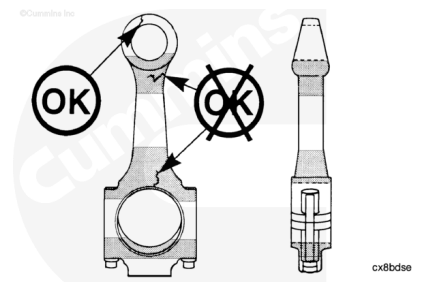
Amperage (Ampere Turns)	
MIN	2600 D.C. or Rectified A.C.
MAX	2800 D.C. or Rectified A.C.

Note : Ampere turn is an electrical current of one ampere flowing through the coil, multiplied by the number of turns in the coil.

The connecting rod **must** be replaced if any indications are visible in the critical (shaded) areas.

⚠ CAUTION ⚠

The rod must be demagnetized completely and cleaned thoroughly. Any small metal particles will cause engine damage.

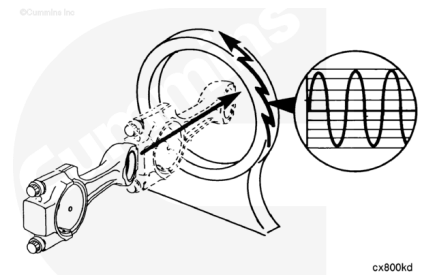


Demagnetize the connecting rod.

Use solvent or steam. Clean the part.

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.